

EXPERIMENT 1:

ELECTROMOTIVE FORCE AND INTERNAL RESISTANCE OF A CELL

Aim: To find the EMF and internal resistance of a cell

Apparatus: dry cell, ammeter, resistance box, and circuit key.

Theory

Let E represent the emf of the cell, R_c its internal resistance, supposed constant R the resistance box reading, R_a the resistance of the ammeter and I the current in the circuit; then

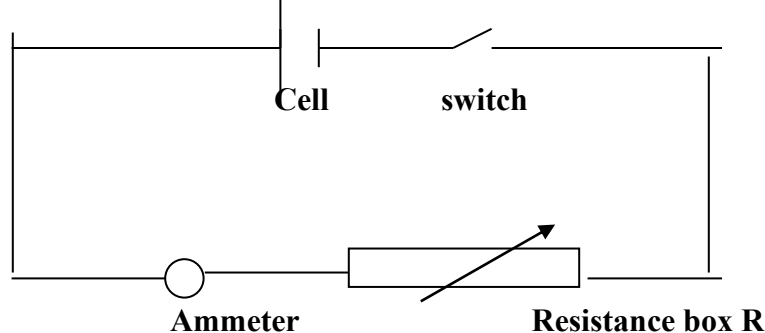
$$\frac{E}{R + R_c + R_a} = I$$

$$\therefore R = E/I - (R_c + R_a)$$

Hence the graph of R against $1/I$ is a straight line whose slope is the magnitude of E and whose negative intercept on the R -axis is the magnitude $(R_c + R_a)$.

Method

Connect the circuit as shown below.



Take $100\ \Omega$ of the resistance box and if the current is measurable on the ammeter scale, record both current and resistance box reading. Gradually reduce the resistance of the box until the current has risen to about 0.1 A . tabulate the results. NB: find out from demonstrator / technician the ammeter resistance value.

Experimental details

1. The current through most resistance boxes should never exceed about 0.1 A .
2. Before taking any readings, give all the plugs in the resistance box a half turn to ensure that they are properly in.
3. If the resistance box possesses an infinity plug it may be used instead of a circuit key.
4. Check the ammeter for zero error and if necessary re-set for zero reading by adjusting screw provided

Analysis

Plot a graph with values of R as ordinates (Y-axis) against the corresponding value of $1/I$ as abscissa (X-axis). From the graph, determine emf and internal resistance of the dry cell.

Error and accuracy

From the graph, estimate the likely errors in E and RC from the difference between the values found from chosen line and those obtained from possible straight line drawn through the points.

Conclusion:-